

Sri Lanka Institute of Information Technology



Application Frameworks – SE3040

**Year 3 Semester 1
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Deployment Report Rural Sanitation Inspection and Issue Tracking System (SDG 6)

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1. Introduction

The Rural Sanitation Inspection and Improvement System deployment technique is described in this deployment report. Eco-friendly Development Goal 6, which focuses on assuring clean water and adequate sanitation, is supported by the system's design. The program offers tools for documenting problems, organizing sanitary inspections, and improving foundation. To assure accessibility, scalability, and practical usability, it was created utilizing the MERN stack and implemented on modern cloud platforms.

2. Deployment Architecture

The frontend, backend, and database layers of the system are divided using a MERN-based design. By assuring that every component of the system serves a different purpose, this design improves maintainability and scalability. While business logic and API communication are handled by the backend, user interaction is handled by the frontend. All application data is safely stored in the database. Third-party APIs are also incorporated to increase system performance and expand functionality.

- Frontend (React) – Vercel
- Backend (Node.js + Express) – Render
- Database (MongoDB) – MongoDB Atlas
- Third-party APIs – Cloudinary, Google OAuth, Nodemailer, OpenStreetMap

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3. Deployment Platforms Used

The system's many components were successfully deployed using a variety of cloud platforms. Each platform was chosen according to its advantages and suitability for the technology used. Render facilitates backend API hosting, while Vercel offers quick frontend application deployment. Trustworthy cloud database services are provided by MongoDB Atlas. The system is improved by additional third-party services that offer communication, storage, and authentication features.

- Frontend: Vercel
- Backend: Render
- Database: MongoDB Atlas
- Image Storage: Cloudinary
- Authentication: Google OAuth
- Email Service: Nodemailer
- Geolocation: OpenStreetMap API

4. Frontend Deployment (Vercel)

Vercel was used to connect the front-end application to the GitHub repository. Setting the proper build settings, creating environment variables, and making sure the application compiled successfully were all part of the deployment process. The program was tested after deployment to ensure that all parts, navigation, and API calls were operating as planned. The frontend, which offers a responsive and user-friendly experience, is now available online.

- Pushed frontend code to GitHub
- Imported project into Vercel
- Configured environment variables
- Deployed application

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5. Backend Deployment (Render)

Render was used to deploy the system's backend, enabling smooth connection with GitHub repositories. Configuring build and start commands, adding necessary environment variables, and making sure the server functions properly were all part of the deployment process. Postman was used to test API endpoints following deployment to confirm functionality. Authentication, data processing, and database and third-party service interactions are all effectively managed by the backend.

- Connected GitHub repository
- Configured build and start commands
- Added environment variables
- Deployed backend service

6. Database Deployment (MongoDB Atlas)

The cloud database solution used to safely store application data was MongoDB Atlas. Access credentials were set up and a cluster was created. To facilitate communication between the backend and database, the connection string was added to the variables in the backend environment. MongoDB ensures effective data recovery and administration throughout the system by storing information about users, villages, inspections, and problems.

- Created cluster
- Configured database access
- Added connection string to backend

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7. Third-Party API Integration Deployment

To improve the system's functionality, several third-party APIs were integrated. Google OAuth allows user identification with Google accounts, while Cloudinary is used for safe picture uploads and storage. OpenStreetMap offers geolocation capabilities, while Nodemailer is used to deliver email alerts. By utilizing dependable outside services, these integrations enhance functionality, user experience, and overall system performance.

- Cloudinary – Image upload
- Google OAuth – Authentication
- Nodemailer – Email notifications
- OpenStreetMap – Geolocation

8. Live URLs

Following successful deployment, public URLs can be used to access the system. Users and evaluators can engage with the application and test its features using these URLs. The backend manages API queries, while the frontend creates the user interface. These links show that the system is operating in a real-world setting and fully implemented.

- Frontend URL: <https://san-track-frontend.vercel.app/>
- Backend URL: <https://santrack-backend.onrender.com>

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9. Challenges Faced

CORS problems, improper environment variable setups, and OAuth redirect mismatches were among the difficulties that arose during the deployment process. It took thorough investigation and configuration parameter tweaks to debug these problems. Managing API integration problems and assuring seamless front-end and back-end interactions presented additional difficulties.

Through testing and debugging, these issues were successfully fixed.

- CORS issues
- Environment variable errors
- OAuth redirect mismatch
- Cloudinary configuration issues

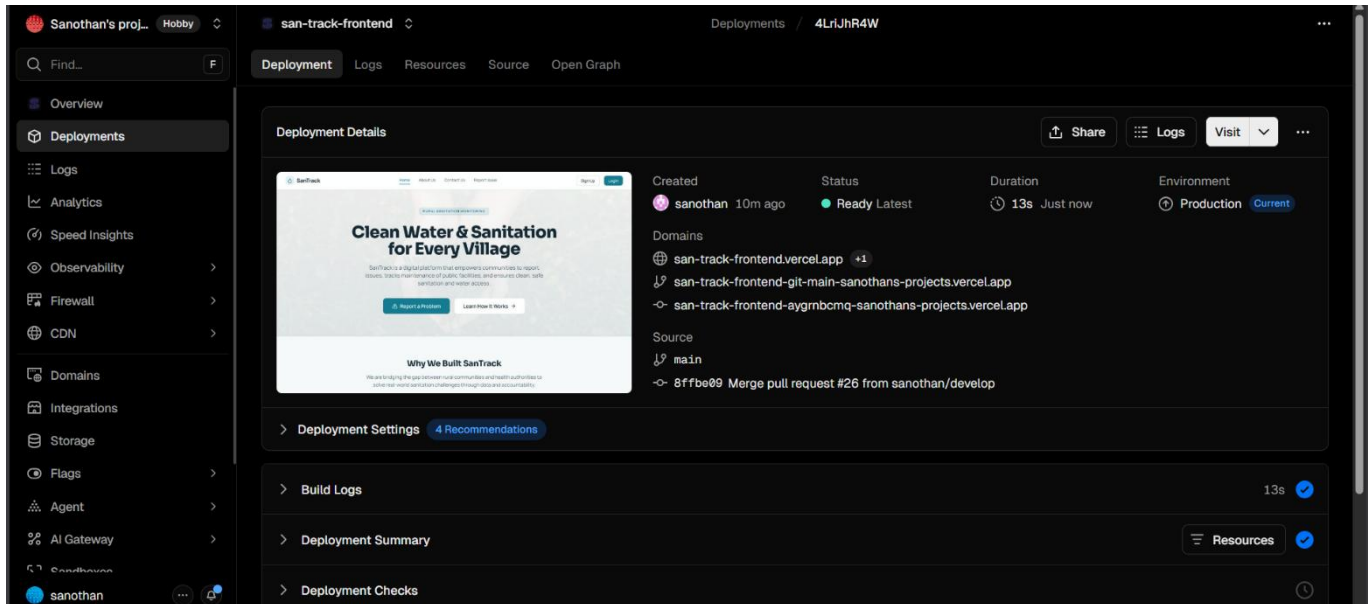
10. Conclusion

Using modern cloud platforms, the Rural Sanitation Inspection and Improvement System was effectively deployed. The system is now fully operational and available online. It shows how the MERN stack and third-party integrations may be used practically. By enhancing sanitation management and monitoring, this initiative contributes to SDG 6. It can be improved in the future with new features.

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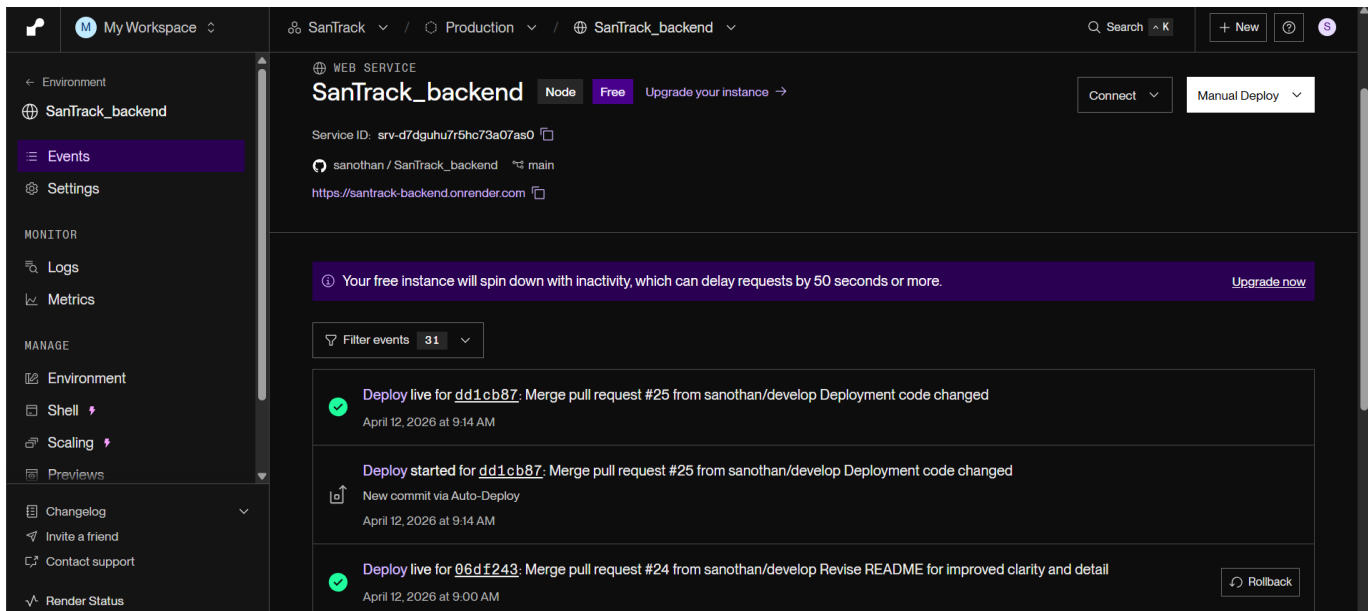
References

Frontend deployment



The screenshot shows the Vercel deployment interface for a project named 'san-track-frontend'. The left sidebar contains navigation options: Overview, Deployments (selected), Logs, Analytics, Speed Insights, Observability, Firewall, CDN, Domains, Integrations, Storage, Flags, Agent, AI Gateway, and Sanitization. The main panel displays 'Deployment Details' for a deployment created 10m ago by 'sanathan'. The status is 'Ready Latest' with a duration of 13s. The environment is 'Production' and 'Current'. The deployment includes domains like 'san-track-frontend.vercel.app' and source code from a 'main' branch. Below the details, there are sections for 'Deployment Settings' (4 Recommendations), 'Build Logs', 'Deployment Summary', and 'Deployment Checks'.

Backend deployment



The screenshot shows the SanTrack backend deployment interface. The left sidebar includes 'My Workspace', 'Environment' (SanTrack_backend), 'Events' (selected), 'Settings', 'MONITOR' (Logs, Metrics), and 'MANAGE' (Environment, Shell, Scaling, Previews, Changelog, Invite a friend, Contact support, Render Status). The main panel displays 'SanTrack_backend' as a 'WEB SERVICE' with a 'Node' runtime and 'Free' plan. It shows the service ID, repository, and main branch. A notification states: 'Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more. Upgrade now'. Below this, a list of deployment events is shown, including 'Deploy live for dd1cb87: Merge pull request #25 from sanathan/develop' and 'Deploy started for dd1cb87: Merge pull request #25 from sanathan/develop'. A 'Rollback' button is visible for the most recent deployment.

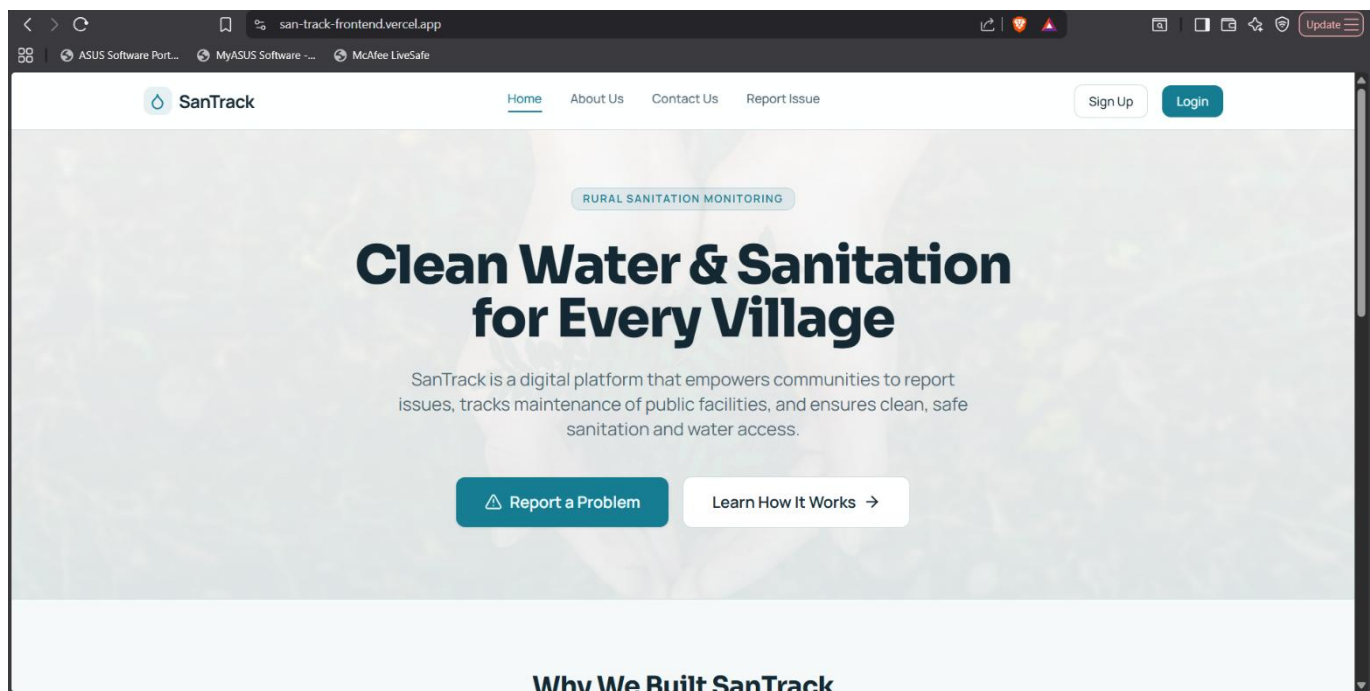
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MongoDB connection

```
All logs ▾  🔍 Search  🕒 Apr 12, 9:13 AM - 9:15 AM  🌐 GMT+5:30  ↻  ⋮

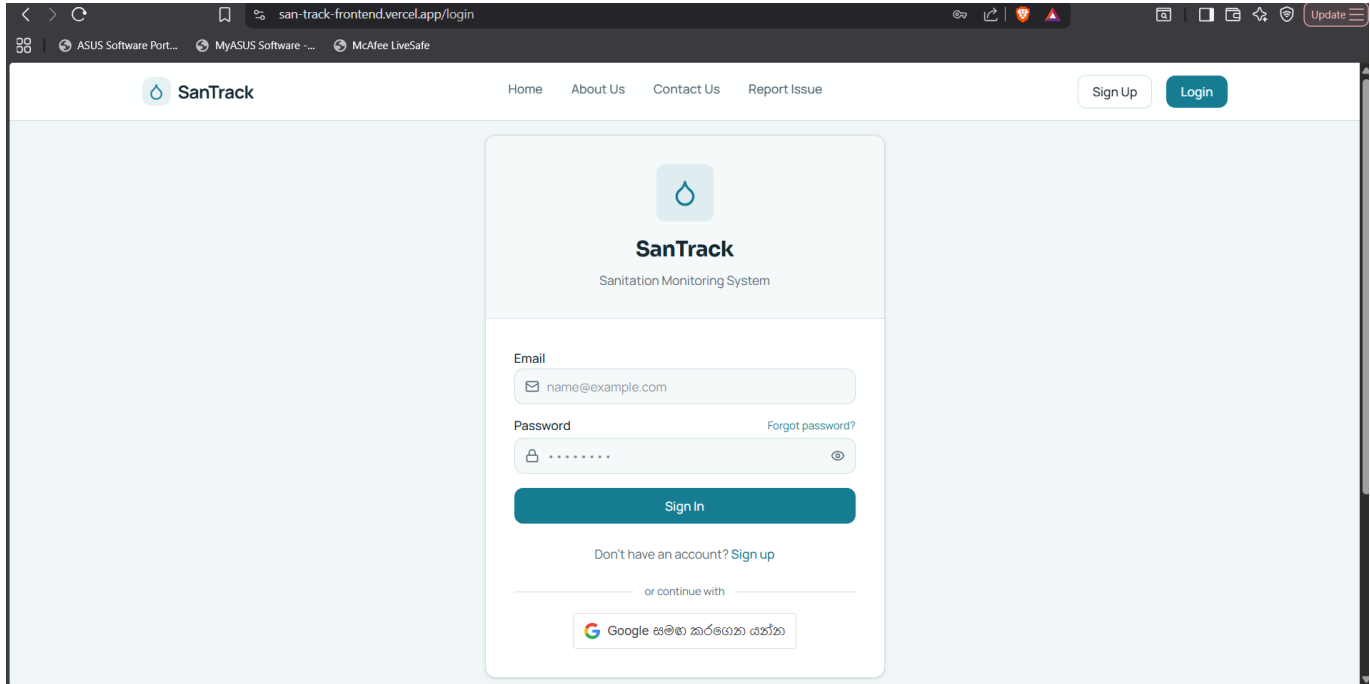
Apr 12
09:14:43 AM [qb41c] [nodemon] to restart at any time, enter `rs`
09:14:43 AM [qb41c] [nodemon] watching path(s): *.*
09:14:43 AM [qb41c] [nodemon] watching extensions: js,mjs,cjs,json
09:14:43 AM [qb41c] [nodemon] starting `node src/server.js`
09:14:43 AM [qb41c] [dotenv@17.3.1] injecting env (0) from .env -- tip: 🗝 agent secret storage: https://dotenvx.com/as2
09:14:47 AM [qb41c] [dotenv@17.3.1] injecting env (0) from .env -- tip: 🔒 encrypt with Dotenvx: https://dotenvx.com
09:14:47 AM [qb41c] Nodemailer credentials missing. Email service will be disabled.
09:14:50 AM [qb41c] MongoDB connected
09:14:50 AM [qb41c] SanTrack server running locally on http://localhost:5000
09:14:54 AM ==> Your service is live 🎉
09:14:54 AM ==>
09:14:54 AM ==> //////////////////////////////////////
09:14:54 AM ==>
09:14:54 AM ==> Available at your primary URL https://santrack-backend.onrender.com
09:14:54 AM ==>
09:14:54 AM ==> //////////////////////////////////////
```

Home page in live URL



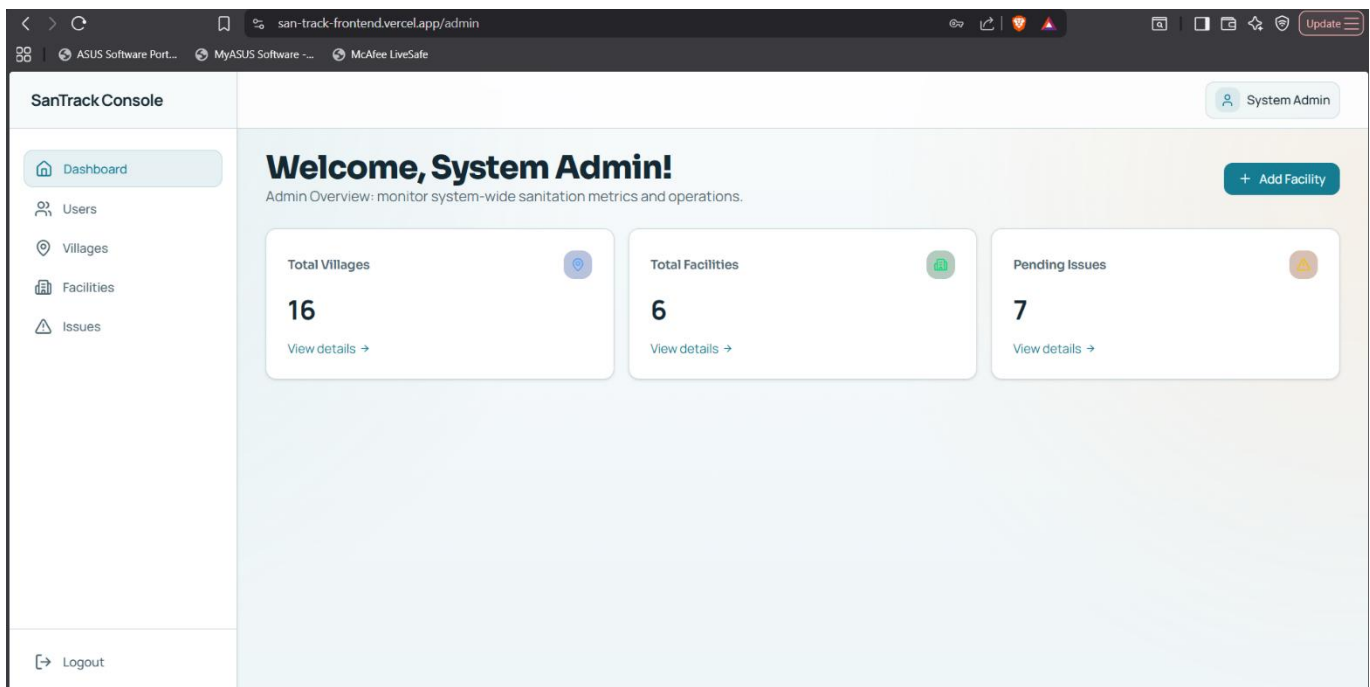
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Login in live URL



The image shows the login page of the SanTrack Sanitation Monitoring System. The page has a dark header with the SanTrack logo and navigation links: Home, About Us, Contact Us, and Report Issue. There are 'Sign Up' and 'Login' buttons in the top right. The main content area features a central login form with a SanTrack logo and the text 'Sanitation Monitoring System'. The form includes fields for 'Email' (with the placeholder 'name@example.com') and 'Password' (with a 'Forgot password?' link). Below the password field is a 'Sign In' button. Underneath the button, there is a link for 'Don't have an account? Sign up' and a section for 'or continue with' which includes a Google login option.

Dashboard in live URL



The image shows the admin dashboard of the SanTrack system. The page has a dark header with the SanTrack logo and navigation links: Home, About Us, Contact Us, and Report Issue. There are 'Sign Up' and 'Login' buttons in the top right. The main content area features a central dashboard with a 'Welcome, System Admin!' message and the text 'Admin Overview: monitor system-wide sanitation metrics and operations.' Below this, there are three cards showing key metrics: 'Total Villages' (16), 'Total Facilities' (6), and 'Pending Issues' (7). Each card has a 'View details' link. On the left side, there is a sidebar with a 'SanTrack Console' header and a list of navigation links: Dashboard, Users, Villages, Facilities, and Issues. At the bottom of the sidebar is a 'Logout' button. In the top right corner of the dashboard, there is a 'System Admin' user profile and an 'Add Facility' button.